

Operator Discovery Report: acceptance_bounded_worse_with_stagnation_safeguard

- Operator type: acceptance.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic bounded-worse acceptance that switches from improving-only to a small threshold/annealed allowance after stagnation, helping escape 2-opt traps without destabilizing later-stage refinement.
- Novelty classification: threshold_acceptance.
- Rediscovery signature: acceptance:threshold:thr=0.016:temp=0.000.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- Use threshold acceptance when a restart solution is not immediately improving.
- Relax or tighten acceptance depending on stagnation and late-search cooling.
- Reject candidate restarts that exceed the configured worse-move budget.

Complexity

- Non-empty lines: 11.
- Complexity score: 0.42.
- Fingerprint: ed4eb3763dec582e.

Why It Should Help

- The operator only changes restart acceptance, so any gain is attributable to a narrow exploration-vs-stability decision rule.

Failure Cases

- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 50.323929, distance-eval delta 0.0.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 890.777714, distance-eval delta 0.0.
- random_restart_2opt gap delta 0.0, runtime delta 14.859786, distance-eval delta 1369.142857.
- sparse_three_opt gap delta 0.0, runtime delta 2.476943, distance-eval delta 0.0.
- clustered_local_search gap delta 0.006181, runtime delta 13.412943, distance-eval delta 1207.428571.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.0.
- Runtime inflation: -0.013835.
- Distance-evaluation inflation: 0.0.
- Same-gap-faster flag: True.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.0.
- improving_only_acceptance mean gap 0.121317 and delta vs full operator 0.0.
- tight_threshold_acceptance mean gap 0.121317 and delta vs full operator 0.0.